

Combinatorics PhD Exam, May 2021

1. Let a_n be the number of ways to select a permutation of length n , and then color each of its cycles red, blue, or green. Find an explicit formula for a_n .
2. Let A be the graph obtained from K_n by deleting an edge. Find a closed formula (so no summation signs) for the number of spanning trees of A .
3. Prove bijectively that the number of graphs with vertex set $[n]$ for which all vertices have even degree is $2^{\binom{n-1}{2}}$.
4. Prove that for every n , there is a tournament T on n vertices with at least $n!/2^{n-1}$ Hamiltonian paths.
5. Let $\mathbb{P} = \{1, 2, \dots\}$ and $(\mathbb{P}, |)$ be the division poset (so $m | n$ if $n/m \in \mathbb{P}$). Find a formula for the Möbius function μ and prove that it is correct. (Hint: decompose intervals in $(\mathbb{P}, |)$ as products of chains.)
6. Let $\mathcal{A}$ be the class of compositions into odd parts, where the size of $\alpha = (a_1, \dots, a_n) \in \mathcal{A}$ is $|\alpha| = a_1 + \dots + a_n$. Find the associated ordinary generating function $A(x) = \sum_{n \geq 0} a_n x^n$ and compute its growth rate $\limsup a_n^{1/n}$.
7. Let $\mathcal{P}$ be the class of integer partitions. Give a bijective proof of the Cauchy identity

$$\prod_{i,j \geq 1} (1 - x_i y_j)^{-1} = \sum_{\lambda \in \mathcal{P}} s_{\lambda}(x_1, x_2, \dots) s_{\lambda}(y_1, y_2, \dots)$$

where s_{λ} is the Schur function of shape λ .

8. Let $\mathcal{A}$ and $\mathcal{B}$ be labeled combinatorial classes with size functions $|\cdot|_{\mathcal{A}}$ and $|\cdot|_{\mathcal{B}}$, respectively. Define an admissible construction $\star$ so that $\mathcal{C} = \mathcal{A} \star \mathcal{B}$ satisfies the exponential generating function identity $C_E(x) = A_E(x) \cdot B_E(x)$.